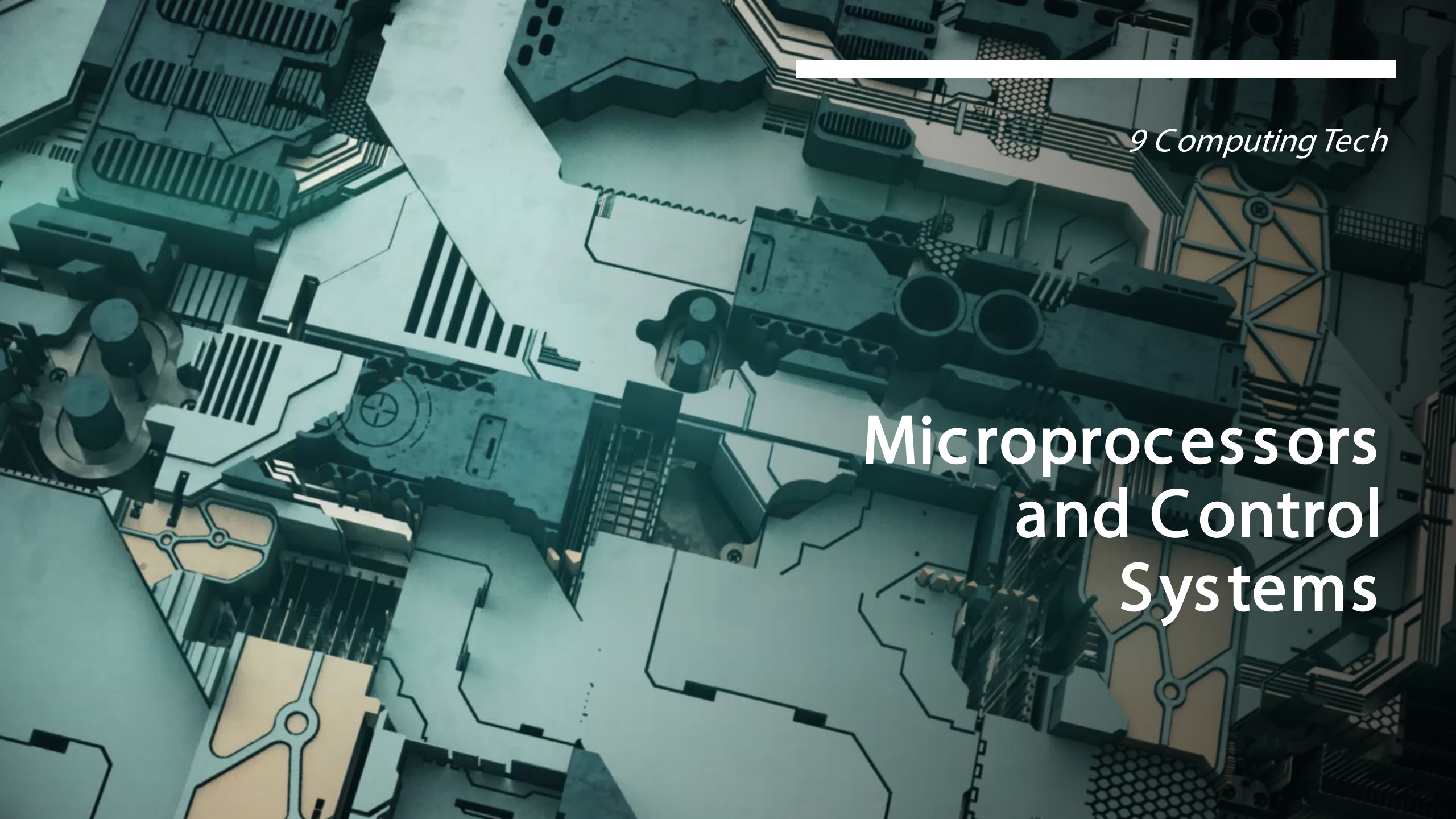


Microprocessors and Control Systems

Learning Intention



Identify mechatronic and automated systems that operate through human or autonomous control



Describe inputs, storage, transmission, processes and outputs in mechatronic and automated systems

Success Criteria



I can distinguish between closed loop and open loop control systems.



I can explain how a real-world control system might function.



I can outline the function of a microprocessor.

Microprocessors

Microprocessor - any of a type of miniature electronic device that contains the arithmetic, logic, and control circuitry necessary to perform the functions of a digital computer's central processing unit. In effect, this kind of integrated circuit can interpret and execute program instructions as well as handle arithmetic operations.



"Brain" of a mechatronic system



Processes input from sensors



Sends signals to actuators



Controls overall system behaviour

Microcontrollers (e.g. Arduino, PICO)

Microcontroller - A small computer built for the purpose of dealing with specific tasks, such as managing the engine in a car, displaying information in a microwave control panel or receiving information from a television remote control. Microcontrollers process data inputted by users (e.g. keypad) or sensors (e.g. light).

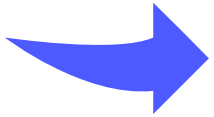




Small, dedicated computers on a board



Used to control automated systems



Accept input → process → produce output



Common in embedded systems

Digital vs Analogue Pins



Digital output: ON/OFF (3V or 0V)



Digital input: detects ON/OFF signals



Analogue output: variable via PWM (Pulse Width Modulation)



Analogue input: reads changing voltages

Control Systems

Control System - a system or a set of devices that manages commands and directs the behaviour of other devices or systems



Manage how a system behaves



Use input → process → output model



Found in automation and robotics



Aim to achieve a desired outcome

Control Loop

Control Loop - a series of control operations, including measuring an output, establishing what the output should be, and taking action to correct it



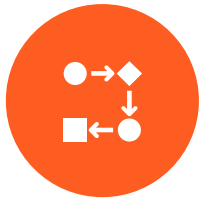
A control loop describes how a system **monitors and adjusts its output**



Involves: **input → process → output → feedback (optional)**



Open-loop: no feedback, output not adjusted



Closed-loop: uses feedback to correct and refine output



Feedback improves **accuracy, stability, and reliability**



Example: thermostat adjusting temperature based on sensor data

Open-Loop Systems

Open-Loop Control System - a type of control system in which the system's output does not have any effect on its input and, therefore, does not have any feedback.

- No feedback used
- Output not checked or adjusted
- Simple and low cost
- Example: toaster or griller

Closed-Loop Systems

Closed-Loop Control System - a control system that uses feedback to adjust its process based on the actual output without human interaction



- Use **feedback** to adjust output
- Output influences input
- More accurate and reliable
- Example: thermostat-controlled air conditioning

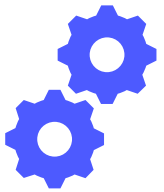
Feedback in Systems



Sensors monitor output



Data sent back to controller



System adjusts automatically



Improves accuracy and stability



Example – Gimbal System

Uses accelerometer (sensor)

Detects tilt (input)

Microcontroller processes data

Servo (actuator) corrects position

Automated Control Systems

Automated Control System - a set of devices and programs designed to manage, command, direct, or regulate the behaviour of other devices or systems without constant human intervention.

Operate without human input

Use sensors, controllers, actuators

Types:

Continuous (24/7 production)

Batch (reprogrammable)

Discrete (small-scale production)

Example: automatic doors, climate control